Required:

Introduction (Aron and Enzo)

Mathematical Model (Complete- Krishiv)

Code/program (Alex and Abi)

Analysis (Manu, Lucas, Gabi)

Other methods/ways to improve (Alex, Krishiv, Diego)

Conclusion (Marco and Gabi)

Video

USE TIMES NEW ROMAN SIZE 12 FOR ALL MAIN BODY TEXTS, AND HEADING 1 FOR ANY TITLES, AND HEADING 2 FOR SUBTITLES

Introduction (Enzo and Aron)

Introduction of variables and model

To model the theoretical patient capacity at a hospital, we have elected to use the following variables:

$$d = number\ of\ doctors$$

$$n = number\ of\ nurses$$

$$e = number\ of\ equipment\ ^{*}(bundled)$$

$$b = number\ of\ beds$$

$$p = number\ of\ patients$$

Since hospitals operate continuously (24/7), staffing variables must account for 3 daily shifts. Therefore:

$d_p = \frac{3}{p_d}$, where p_d is patients per doctor per shift

$n_p = \frac{3}{p_n}$, where p_n is patients per nurse per shift

Based on these variables, we have developed the following constraints:

$$Base\ constraint: P \leq (\frac{d}{d_p}, \frac{n}{n_p}, \frac{e}{e_p}, \frac{b}{b_p})$$

where x_p refers to the quantity of variables per patient

$$Budget\ Constraint: C_d \cdot d + C_n \cdot n + C_e \cdot e + C_b \cdot b \leq B$$

where C_x refers to the cost of variable x and B refers to overall hospital budget

*: Bundled equipment refers to a combination of equipment used in hospitals, such as, monitors, ventilators, imaging systems…

$$Doctor\ Minimum\ Staffing: d \geq d_{min}$$

$$Nurse\ Minimum\ Staffing: n \geq n_{\min}$$

where $d_{\min}$and $n_{\min}$represent regulatory minimum staffing requirements.

$$Area\ Constraint: A_e \cdot e + A_b \cdot b \leq A_T$$

where A_x refers to the area taken by variable x and A_T refers to total area available

$$Non-negativity: (d, n, e, b, p) \geq 0, \qquad \mathrm{d, n, e, b} \in \mathrm{Z}$$

Development and manipulation of the model

$$f = \max(p)$$

Since p is being maximized, we can re-write the base constraints in the form:

$$d = d_p \cdot p, \quad n = n_p \cdot p, \quad e = e_p \cdot p, \quad b = b_p \cdot p$$

We can substitute these values into the budget constraint:

$$C_d \cdot (d_p \cdot p) + C_n \cdot (n_p \cdot p) + C_e \cdot (e_p \cdot p) + C_b \cdot (b_p \cdot p) \leq B$$

We can re-arrange the equation to isolate p:

$$p(C_d \cdot d_p + C_n \cdot n_p + C_e \cdot e_p + C_b \cdot b_p) \leq B$$

$$p \leq \frac{B}{(C_d \cdot d_p + C_n \cdot n_p + C_e \cdot e_p + C_b \cdot b_p)}$$

We can repeat the steps to write the area constraint in terms of p:

$$A_e \cdot (e_p \cdot p) + A_b \cdot (b_p \cdot p) \leq A_T$$

$$p(A_e \cdot e_p + A_b \cdot b_p) \leq A_T$$

$$p \leq \frac{A_T}{(A_e \cdot e_p + A_b \cdot b_p)}$$

Hence, we can write our final model as:

$$p \leq \min\left(\frac{B}{(C_d \cdot d_p + C_n \cdot n_p + C_e \cdot e_p + C_b \cdot b_p)}, \frac{A_T}{(A_e \cdot e_p + A_b \cdot b_p)}\right)$$

With the feasibility constraint:

$$n \geq p \cdot K$$

$$n_p \cdot p \geq p \cdot K$$

$$n_p \geq K$$

And the non-negativity constraints:

$$(d, n, e, b, p) \geq 0$$

This model, alongside its related constraints, can be developed into a model on Python that can solve for the maximum capacity a hospital can maintain with a set of given variables.

Code (Alex and Abi)

Analysis (Manu, Lucas, Gabriel)

Areas to improve (Alex, Krishiv, Diego)

Areas to improve mathematical modelling

The model developed represents the fundamental resources employed in a hospital and provides rudimentary constraints for each resource. In reality, hospitals maintain a much larger number of resources, such as specified instruments available, operation theatres… This requires unique constraints for each resource, meaning that to truly optimize the resources a hospital has available, there would be a large increase in the number of resources and constraints employed. Our model simplified the resources, and their respective constraints, and used simple constraints.

Conclusion (Gabriel and Marco)